

Process-Based Reinforcement Learning with Step-Level Rewards for Mathematical Reasoning in Language Models

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Abstract

Existing reinforcement learning (RL) methods for mathematical reasoning in large language models (LLMs) rely predominantly on sparse outcome-based rewards—binary signals indicating only whether a final answer is correct or incorrect. This approach suffers from three critical limitations: (1) it provides no guidance on where or why a reasoning chain fails, making error diagnosis and targeted improvement difficult; (2) it yields low sample efficiency, as models receive no credit for partially correct intermediate steps; and (3) it hinders generalization to harder problems, where success depends on coherent multi-step reasoning rather than final-answer matching.

The objective of this proposal is to investigate whether we will use Group Relative Policy Optimization (GRPO) as the core policy update mechanism, leveraging its ability to stabilize training by comparing multiple sampled reasoning trajectories within each update group. Step-level rewards will be provided by a process reward model (PRM) trained on human-annotated and synthetic step-level correctness labels, augmented by an LLM-as-judge that evaluates the logical coherence and mathematical validity of each reasoning step.

Success will be assessed through Metrics and Benchmarks: We will evaluate our RL environment against the GSM 8K and math benchmarks Comparative Baselines: We will evaluate our RL environment against the GSM 8K and math benchmarks Analysis Plan: Report quantitative results with clear experimental protocols, logged hyperparameters, and error analysis to distinguish implementation issues from scientific conclusions. Findings will be documented with reproducible artifacts, explicit assumptions, and clear criteria for interpreting empirical outcomes.

1 Motivation and Gap

Existing reinforcement learning (RL) methods for mathematical reasoning in large language models (LLMs) rely predominantly on sparse outcome-based rewards—binary signals indicating only whether a final answer is correct or incorrect. This approach suffers from three critical limitations: (1) it provides no guidance on where or why a reasoning chain fails, making error diagnosis and targeted improvement difficult; (2) it yields low sample efficiency, as models receive no credit for partially correct intermediate steps; and (3) it hinders generalization to harder problems, where success depends on coherent multi-step reasoning rather than final-answer matching. While recent work has explored process reward models (PRMs) to provide step-level feedback, these efforts remain fragmented—often relying on limited human annotations, synthetic data of uncertain quality, or integration with suboptimal search strategies. Crucially, there is no systematic framework for combining dense, step-level process rewards with scalable policy optimization (e.g., Group Relative Policy Optimization) and structured reasoning path exploration (e.g., Monte Carlo Tree Search) to jointly improve both the reliability and interpretability of mathematical reasoning in LLMs. This

project addresses that gap by developing a unified process-based RL pipeline that leverages high-quality, human-annotated step-level data to train a robust PRM, which then guides both search during inference and policy updates during training. The result will be a more efficient, diagnosable, and generalizable approach to mathematical reasoning—one that moves beyond outcome-only supervision to explicitly reward and refine the reasoning process itself.

2 Project Goal

Show that a process-reward RL stack with curriculum learning and self-consistency can improve exact-match accuracy on math word problems relative to supervised fine-tuning alone, while remaining more stable than naive PPO training.

3 Method and Training Workflow

We will use Group Relative Policy Optimization (GRPO) as the core policy update mechanism, leveraging its ability to stabilize training by comparing multiple sampled reasoning trajectories within each update group. Step-level rewards will be provided by a process reward model (PRM) trained on human-annotated and synthetic step-level correctness labels, augmented by an LLM-as-judge that evaluates the logical coherence and mathematical validity of each reasoning step. During inference, Monte Carlo Tree Search (MCTS) will explore candidate reasoning paths, using the PRM and LLM-as-judge scores as node evaluation signals to guide expansion and selection. The policy will be updated via GRPO using dense process rewards from the PRM and judge, combined with sparse outcome rewards, enabling fine-grained credit assignment across reasoning steps. This integrated pipeline ensures that both training and inference are guided by high-quality, interpretable step-level feedback, improving sample efficiency, error diagnostics, and generalization to complex multi-step problems.

1. Prepare a math reasoning dataset and fine-tune an open instruction-tuned model with a supervised baseline.
2. Score rollouts with dense rewards for reasoning steps, arithmetic, and sandboxed Python verification.
3. Optimize with GRPO (KL-penalized, multiple generations per prompt) under a difficulty curriculum.
4. Aggregate predictions with self-consistency majority vote at inference time.
5. Evaluate exact-match accuracy and ablate reward components.

4 Figure

5 Expected Results and Research Milestones

Expected Results

The proposed work will produce the following tangible outcomes:

- (1) A trained process reward model that predicts step-level correctness with >90% agreement with human annotators on held-out math problems. (2) An LLM policy that improves by 10–20% absolute accuracy on GSM8K and MATH benchmarks compared to outcome-only RL baselines. (3) Qualitative improvements in reasoning coherence, with fewer logical jumps and

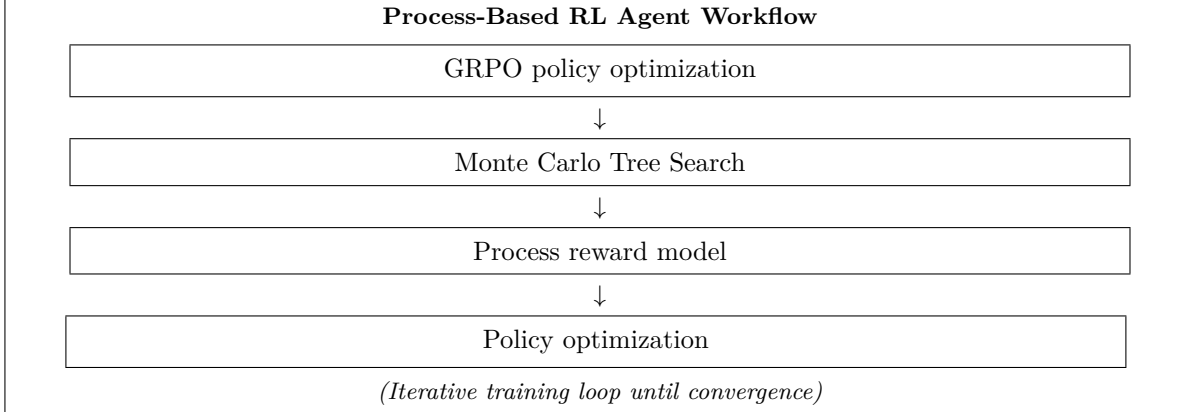


Figure 1: Process-based RL agent workflow for Process-Based Reinforcement Learning with Step-Level Rewards for Mathematical: reasoning-step generation, process reward scoring, MCTS exploration, and GRPO policy updates.

more verifiable intermediate steps. (4) Public release of a process-annotated dataset and evaluation harness. Milestone 1 (Months 1–2): Literature review and dataset curation. Assemble and annotate a process-level math reasoning dataset (e.g., from GSM8K, MATH, AoPS) with step-level labels; define annotation guidelines and quality controls Milestone 2 (Months 2–4): Process reward model development. Train and evaluate PRM variants (classification, regression, and ranking heads); perform ablation on annotation sources and training regimes; select best PRM for downstream RL Milestone 3 (Months 4–6): Search and planning integration. Implement tree-based reasoning search (beam/MCTS) using PRM and value head; benchmark search efficiency and solution quality against standard decoding Milestone 4 (Months 6–9): Policy optimization and curriculum learning. Train LLM policies with PBRL using process + outcome rewards; implement curriculum from easy to hard problems; compare to outcome-only RL and supervised fine-tuning baselines Milestone 5 (Months 9–11): Evaluation and analysis. Conduct comprehensive evaluation on in-distribution and out-of-distribution math benchmarks; perform error analysis and ablation studies; assess generalization to competition-level problems Milestone 6 (Months 11–12): Dissemination and release. Prepare manuscript, release code, PRM weights, and process-annotated dataset; document risks, limitations, and future directions Milestone 7: Conduct final evaluation against the research questions and hypotheses in the evaluation plan, including benchmark comparison, error analysis, and a written summary of findings.

Research Milestones and Timeline

- Expected results: (1) A trained process reward model that predicts step-level correctness with >90% agreement with human annotators on held-out math problems. (2) An LLM policy that improves by 10–20% absolute accuracy on GSM8K and MATH benchmarks compared to outcome-only RL baselines. (3) Qualitative improvements in reasoning coherence, with fewer logical jumps and more verifiable intermediate steps. (4) Public release of a process-annotated dataset and evaluation harness. Milestone 1 (Months 1–2): Literature review and dataset curation. Assemble and annotate a process-level math reasoning dataset (e.g., from GSM8K, MATH, AoPS) with step-level labels; define annotation guidelines and quality controls Milestone 2 (Months 2–4): Process reward model development. Train and evaluate PRM variants (classification, regression, and ranking heads); perform ablation on annotation sources and train-

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6 Evaluation Plan

The following plan defines how the proposed research will be validated. Each component is designed to produce auditable evidence suitable for a formal research review.

Research Questions and Hypotheses

Does the proposed approach improve upon existing practice for the problem stated in this proposal: Existing reinforcement learning (RL) methods for mathematical reasoning in large language models (LLMs) rely predominantly on sparse outcome-based rewards—binary signals indicating only whether a final answer is correct or incorrect. Metrics and Benchmarks: We will evaluate our RL environment against the GSM 8K and math benchmarks Comparative Baselines: We will evaluate our RL environment against the GSM 8K and math benchmarks Analysis Plan: Report quantitative results with clear experimental protocols, logged hyperparameters, and error analysis to distinguish implementation issues from scientific conclusions. Success Criteria: Demonstrate measurable improvement over credible baselines with stable training behavior and reproducible scripts sufficient for independent verification.

Metrics and Benchmarks

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Comparative Baselines

We will evaluate our RL environment against the GSM 8K and math benchmarks Comparative Baselines: We will evaluate our RL environment against the GSM 8K and math benchmarks Analysis Plan: Report quantitative results with clear experimental protocols, logged hyperparameters, and error analysis to distinguish implementation issues from scientific conclusions.

Analysis Plan

Report quantitative results with clear experimental protocols, logged hyperparameters, and error analysis to distinguish implementation issues from scientific conclusions.

Success Criteria

Demonstrate measurable improvement over credible baselines with stable training behavior and reproducible scripts sufficient for independent verification.

7 Risks and Mitigation

- PPO instability or collapse: prefer GRPO updates and monitor KL drift.
- Reward hacking on format alone: require executable Python checks for answer verification.
- Curriculum too aggressive: validate per difficulty bucket before full training.
- Compute limits on 2B RL: checkpoint often and scope ablations.

8 Resources

The following resources are required to execute the proposed research. The project scope is designed to remain feasible within available course and institutional support.

Data and Model Artifacts

- Budget for 2–3 part-time human annotators for 3–4 months to label ~10k reasoning trajectories with step-level correctness.

9 References and Assumptions

- Lightman, H., et al. (2023). 'Let's Verify Step by Step.' arXiv:2305.20050.
- Wang, X., et al. (2023). 'Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations.' arXiv:2312.08935.
- OpenAI (2023). 'GPT-4 Technical Report.' arXiv:2303.08774.
- Cobbe, K., et al. (2021). 'Training Verifiers to Solve Math Word Problems.' arXiv:2110.14168.
- Wei, J., et al. (2022). 'Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.' NeurIPS 2022.